Printed Page: 2 University Roll No.....

First Mid-Term Examination, Odd Semester 2019-20

Program: B-Tech-CSE, Year: II Semester: III

BCSC 0004: Operating Systems

Time: 1 Hour **Maximum Marks: 15**

Note: Attempt All the Questions

Section – A

3 x 2 = 6 Marks

I. Comment in your opinion, which operating system structure modern MS-Windows OS based system follows. Do you know any other type of operating system structure, If Yes, Explain in brief? Also enlist different types of services offered by operating system.

II. Explain System calls with example. Which system call is used to create clone or child process or thread of a process? Define multithreading.

III. Many CPU-scheduling algorithms are parameterized. For example, the Round Robin algorithm requires a parameter to indicate the time slice. Multilevel feedback queues require parameters to define the number of queues, the scheduling algorithms for each queue, the criteria used to move processes between queues, and so on.
These algorithms are thus really sets of algorithms. One set of algorithms may include another (for example, the FCFS algorithm is the RR algorithm with an infinite time quantum). What (if any) relation holds between the following pairs of algorithm sets?

a. Priority and SJF
b. Multilevel feedback queues and FCFS
c. Priority and FCFS
d. RR and SJF

Section – B

3 x 3 = 9 Marks

I. Explain preemption and non preemption strategy. Consider if a process is migrating from running to waiting (blocked) state, is it preemption or not, Justify with suitable reason. What is the role of two suspend states in process transition diagram.

II. Overlapping of CPU time and IO time affects CPU utilization, Comment on the statement with suitable reasons. Define Multiprogramming. What are CPU bound processes and IO bound processes. Consider the following program (assume it to be

syntactically correct) and determine whether it is CPU bound or IO bound with justified reason.

```
main() {
scanf("%d",&a);
scanf("%d",&b);
c=a+b;
printf("%d",&c); }
```

III. Suppose that following processes arrive for execution at the times indicated. Each process will run for the amount of time listed. Assume **Non-Preemptive** scheduling in answering the questions.

Process	Arrival Time	Burst Time (CPU Time)
P1	0.0	8
P2	0.4	4
P3	1.0	1

i) Calculate average turnaround time for these processes by applying SJF scheduling algorithm

ii) Consider a scenario of future-knowledge scheduling algorithm. The SJF algorithm is supposed to improve performance, but notice that we chose to run process P1 at time 0 because we did not know that two shorter processes would arrive soon. Compute the average turnaround time if the CPU is left idle for the first 1 unit and then SJF scheduling is used. Remember that processes P1 and P2 are waiting during this idle time, so their waiting time may increase.

iii) Discuss a scenario where SJF scheduling lead to starvation of a process.

Printed Pages:1 University Roll No.....

First Mid-Term Examination, Odd Semester 2019-20

B.Tech(CSE), II Year, III Semester

BCSC0005: Computer Organization

Time : 1 Hour Maximum Marks: 15

Section – A

Note: All questions are compulsory 3 x 2 = 6 Marks

Q1. Briefly explain the working of carry look-ahead adder by giving its advantage over binary adder.

Q2. Perform the arithmetic operations (+42) + (-13) and (-42) - (-13) by using signed-2's complement representation and show that whether overflow occurred or not. Assume 8-bit registers hold the binary value.

Q3. A digital computer has a common bus system for 16 registers of 32 bits each.

a. How many selection inputs are there in each multiplexer?
b. What size of multiplexers are needed?
c. How many multiplexers are there in the bus?

Section – B

Note: All questions are compulsory 3 x 3 = 9 Marks

Q1. Draw the flowchart to be used for the addition and subtraction of two decimal numbers in signed-magnitude representation.

Q2. Represent the number $(-29.25)_{10}$ by using IEEE-754 32-bit and 64-bit floating point representation.

Q3. Show the step-by-step multiplication process using Booth algorithm when the numbers (+25)& (-13) are multiplied. Assume 6-bit registers hold the signed numbers.

Printed Pages: 02 University Roll No. :- _____

First Mid Term Examination, 2019 - 20
B. Tech. II Year III Semester
Engineering Mathematics III (BMAS 0103)

Time: 1 Hr. **Note:** Attempt **ALL** sections. **Total Marks:** 15

SECTION A **(2 x 3 = 6 marks)**

Instructions: This section has **3** questions. Attempt **ALL** questions. The symbols have their usual meanings.

Q.1. Write down the following partial differential equations in proper form: (0.5 x 4 =2)

(a) Two dimensional heat flow equations in unsteady state

(b) One dimensional heat flow equation

(c) Laplace's equation in cartesian coordinates

(d) Two dimensional equations for vibrating membranes

Q.2. Obtain the Fourier series for the function $f(x) = x^2, -\pi \leq x \leq \pi$. Hence obtain the following sum:

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$$

Q.3. Use the method of separation of variables to solve:

$$\frac{\partial u}{\partial x} = 2\,\frac{\partial u}{\partial t} + u$$

Given that: $u(x, 0) = 6\,e^{-3x}$

SECTION B (3 x 3 = 9 marks)

Instructions: This section has **3** questions. Attempt **ALL** questions. The symbols have their usual meanings.

Q.1. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = A\ sin\frac{\pi x}{l}$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by,

$$y = A\ sin\frac{\pi x}{l} cos\frac{\pi ct}{l}$$

Q.2. Express $f\ (x) = x$ as a

(a) half range sine series in $0 < x < 2$.

(b) half range cosine series in $0 < x < 2$.

Q.3. Classify the following differential equations:

(a) $4\frac{\partial^2 u}{\partial x^2} - 4\frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial t^2} = 0$

(b) $z_{xx} = z_{yy}$

Printed Pages: 1 University Roll No.....

First Mid-Term Examination, Odd Semester 2019-20

B.Tech (CSE/CCV/DA/IoT), IInd Year, IIIrd Semester

BCSC0002 : Object-Oriented Programming

Time : 1 Hour Maximum Marks: 15

Section – A

Three questions of 02 marks each (with no internal choice). 3 x 2 = 6 Marks

I. Describe all the features of Object Oriented Programming.
II. How is Data Abstraction different from Data Hiding.
III. Difference between function Overloading and function Overriding.

Section – B

Three questions of 03 marks each (with no internal choice). 3 x 3 = 9 Marks

I. Explain inheritance and write a program in Java to implement it
II. What do you understand by constructor? How many types of constructor are there? Define the condition for the Java compiler to supply the default constructor.
III. How many types of variables can exist inside a Java program? Discuss the scope of a variable.

SET-A

Printed Pages: 04 University Roll No.....

First Term Examination, Odd Semester 2019-20

B. Tech (All Branches), II Year, III Semester

Subject: English for Professional Purposes–I

Subject Code: BELH-0003

Time: 1 Hour Maximum Marks: 15

Instructions:

1. It is mandatory to clearly mention your set number in the answer sheet.
2. While answering Verbal Aptitude questions, wherever applicable, mention the options (A, B, C etc.) only in the answer sheet.
3. Word limit for the questions in Section A is 60 while for the question in Section B is 150.

PART A - ORAL SKILLS

Section- A **1.5x3 = 4.5 Marks**

Attempt all the questions.

1. Read the passage given below and decide which level of communication does the passage talk about? Give two reasons.
   When stuck in traffic, you may think.....
   "I'm always late ... I don't know what does the guy going to think?... I should have got my stuff ready so that I could leave early...why am I like that?"...
   Now observe that your thoughts are not related to traffic anymore it's all about only you.
   Suddenly, you see a sticker on the car which is in front of you and sticker says 'physically challenged, be patient'.
   Now what would be your changed thoughts?
   I am blessed that I am healthy... I can wait 5 more minutes... my friend can wait if he is genuine...it's okay to be late, I am nice at heart...
   This happens every single moment just the scenarios are different.

2. See the cartoon and decide what kind of barrier it will createfor a Hindi Speaker who knows that widows wear white dress in Hindi Speaking Belt of India, but does not know anything about Christian culture. Elucidate.

3. Do you agree with the statement, 'Verbal and non-verbal signals should go hand in hand'? Why or why not?

Section- B

3x1 = 3 Marks

Attempt any one question.

4. Explain the process of communication with the help of diagram.
5. Rewrite the following sentences in order to make them more appropriate for professional communication.
   a) Our manager is more positively optimistic than was the case last week.
   b) The old widow woman asked the solicitor to draw up her will.
   c) He dropped out of school on account of the fact that it was necessary for him to help support his family.
   d) I saw a man yesterday whose beard was long.
   e) He is a decent college graduate.
   f) Are you going to eat Raj.

PART B- VERBAL APTITUDE

Attempt all the questions. Each question carries 0.25 mark.

Identify the part of speech of the words written in bold and italics. (Q 1-5)

1. The rumour of Professor Agrawal's appointment as Chief Secretary of the State spread very ***fast***.
   A. Noun B. Pronoun
   C. Adverb D. Adjective
   E. Verb
2. There's ***nothing*** in the drawer. I took everything out.
   A. Noun B. Pronoun
   C. Adverb D. Adjective
   E. Verb
3. I tried ***on*** a few jackets, but none of them looked nice.
   A. Noun B. Pronoun
   C. Verb D. Preposition
   E. Adverb

4. He has a ***commanding*** lead in the championships.
   A. Verb B. Pronoun
   C. Adverb D. Adjective
   E. Noun
5. It's the third in a sequence of three books, the first of ***which*** I really enjoyed.
   A. Pronoun B. Noun
   C. Verb D. Preposition
   E. Adverb

Give one example each of the following (Q 6-10):

6. Split infinitive
7. A passive sentence beginning with 'Let'
8. 'Up' as an adverb
9. Linking verb
10. Anaphoric 'to'

Choose the most appropriate option as your answer. (Q 11-23)

11. It is three years since I last _____ Rohan.
    A. had seen B. has seen
    C. saw D. see
    E. None of the above
12. I _____ Raj this afternoon. (time of speaking – 3 p.m.)
    A. didn't see B. have not seen
    C. had not seen D. will see
    E. None of the above
13. Jewellery of worth rupees thirty thousand has been stolen. The thieves_____ the shop by cutting a hole in the ceiling.
    A. had entered B. have entered
    C. entered D. had been entered
    E. None of the above
14. She ___ in Mathura for a long time but now she lives in Noida.
    A. lived B. has lived
    C. lives D. will live
    E. None of the above
15. He asked me to teach Mathematics. But it was ten years since I _____ any lecture on the subject.
    A. had given B. has given
    C. gave D. have given
    E. None of the above
16. It __ quite often in Britain during the winter.
    A. is snowing B. has snowed
    C. has been snowing D. snows
    E. None of the above
17. Which of the following is not the non-finite form of a verb?
    A. Infinitive B. Present Participle
    C. Modal D. Gerund
    E. Past Participle
18. It is not so easy to cook as _____.
    A. eats B. eat
    C. eating D. to eat
    E. None of the above
19. The teenager refused _____on holiday with his parents.
    A. go B. going (present participle)
    C. going (gerund) D. to go
    E. went
20. Identify the underlined word in: 'Can you help me fix the leaning column of blocks?
    A. Finite Verb B. Past Participle
    C. Gerund D. Present Participle
    E. Infinitive
21. Why do you make your parents _____so much about you?
    A. worry B. worried
    C. to worry D. worrying

E. None of the above

22. I want to prepare and ____ the sandwich now.
A. eat B. to eat
C. eating D. ate
E. None of the above

23. When you hear through the grapevine, you _____.
A. are intoxicated
B. are using hearing machine
C. learn about it informally
D. are informed about it through proper channel
E. None of these

Choose the correct form of verb. (Q 24-30)

Fiat was started by a group of Italian businessmen in 1899. In 1903, Fiat, ___24__ 132 cars. Some of these cars _25__by the company to the United States and Britain. In 1920, Fiat___26___ making cars at a new factory at Lingotto, near Turin. There was a track on the roof where the cars____27____ by technicians. In 1936, Fiat launched the Fiat 500. This car ___28___the Topolino – the Italian name for Mickey Mouse. The company grew, and in 1963 Fiat ___29___more than 300,000 vehicles. Today, Fiat is based in Turin, and its cars ___30___ all over the world.

24. A. produced B. was produced
25. A. exported B. were exported
26. A. started B. was started
27. A. tested B. were tested
28. A. called B. was called
29. A. exported B. was exported
30. A. sold B. are sold

SET-B

Printed Pages: 04 **University Roll No.....**

First Term Examination, Odd Semester 2019-20

B. Tech (All Branches), II Year, III Semester

Subject: English for Professional Purposes–I

Subject Code: BELH-0003

Time: 1 Hour **Maximum Marks: 15**

Instructions:

1. It is mandatory to clearly mention your set number in the answer sheet.
2. While answering Verbal Aptitude questions, wherever applicable, mention the options (A, B, C etc.) only in the answer sheet.
3. Word limit for the questions in Section A is 60 while for the question in Section B is 150.

PART A - ORAL SKILLS

Section- A **1.5x3 = 4.5 Marks**

Attempt all the questions.

1. Read the passage given below and decide which level of communication does the passage talk about? Give two reasons.

   When stuck in traffic, you may think.....

   "I'm always late ... I don't know what does the guy going to think?... I should have got my stuff ready so that I could leave early...why am I like that?"...

   Now observe that your thoughts are not related to traffic anymore it's all about only you.

   Suddenly, you see a sticker on the car which is in front of you and sticker says 'physically challenged, be patient'.

   Now what would be your changed thoughts?

   I am blessed that I am healthy... I can wait 5 more minutes... my friend can wait if he is genuine...it's okay to be late, I am nice at heart...

   This happens every single moment just the scenarios are different.

2. See the cartoon and decide what kind of barrier it will createfor a Hindi Speaker who knows that widows wear white dress in Hindi Speaking Belt of India, but does not know anything about Christian culture. Elucidate.

3. Do you agree with the statement. 'Verbal and non-verbal signals should go hand in hand'? Why or why not?

Section- B **3x1 = 3 Marks**

Attempt any one question.

4. Explain the process of communication with the help of diagram.
5. Rewrite the following sentences in order to make them more appropriate for professional communication.
   a) Our manager is more positively optimistic than was the case last week.
   b) The old widow woman asked the solicitor to draw up her will.
   c) He dropped out of school on account of the fact that it was necessary for him to help support his family.
   d) I saw a man yesterday whose beard was long.
   e) He is a decent college graduate.
   f) Are you going to eat Raj.

PART B- VERBAL APTITUDE

Attempt all the questions. Each question carries 0.25 mark.

Choose the most appropriate option as your answer. (Q 1-13)

1. It is three years since I last _____ Rohan.
   A. had seen B. has seen
   C. saw D. see
   E. None of the above
2. I _____ Raj this afternoon. (time of speaking – 3 p.m.)
   A. didn't see B. have not seen
   C. had not seen D. will see
   E. None of the above
3. Jewellery of worth rupees thirty thousand has been stolen. The thieves_____ the shop by cutting a hole in the ceiling.
   A. had entered B. have entered
   C. entered D. had been entered
   E. None of the above
4. She ___ in Mathura for a long time but now she lives in Noida.
   A. lived B. has lived

C. lives D. will live
E. None of the above

5. He asked me to teach Mathematics. But it was ten years since I _____ any lecture on the subject.
A. had given B. has given
C. gave D. have given
E. None of the above

6. It __ quite often in Britain during the winter.
A. is snowing B. has snowed
C. has been snowing D. snows
E. None of the above

7. Which of the following is not the non-finite form of a verb?
A. Infinitive B. Present Participle
C. Modal D. Gerund
E. Past Participle

8. It is not so easy to cook as _____.
A. eats B. eat
C. eating D. to eat
E. None of the above

9. The teenager refused _____on holiday with his parents.
A. go B. going (present participle)
C. going (gerund) D. to go
E. went

10. Identify the underlined word in:
'Can you help me fix the leaning column of blocks?
A. Finite Verb B. Past Participle
C. Gerund D. Present Participle
E. Infinitive

11. Why do you make your parents _____so much about you?
A. worry B. worried
C. to worry D. worrying
E. None of the above

12. I want to prepare and ____ the sandwich now.
A. eat B. to eat
C. eating D. ate
E. None of the above

13. When you hear through the grapevine, you _____.
A. are intoxicated
B. are using hearing machine
C. learn about it informally
D. are informed about it through proper channel
E. None of these

Identify the part of speech of the words written in bold and italics. (Q 14-18)

14. The rumour of Professor Agrawal's appointment as Chief Secretary of the State spread very ***fast***.
A. Noun B. Pronoun
C. Adverb D. Adjective
E. Verb

15. There's ***nothing*** in the drawer. I took everything out.
A. Noun B. Pronoun
C. Adverb D. Adjective
E. Verb

16. I tried ***on*** a few jackets, but none of them looked nice.
A. Noun B. Pronoun
C. Verb D. Preposition
E. Adverb

17. He has a ***commanding*** lead in the championships.
A. Verb B. Pronoun
C. Adverb D. Adjective
E. Noun

18. It's the third in a sequence of three books, the first of ***which*** I really enjoyed.
A. Pronoun B. Noun

C. Verb D. Preposition
E. Adverb

Choose the correct form of verb. (Q 19-25)

Fiat was started by a group of Italian businessmen in 1899. In 1903, Fiat, ___19__ 132 cars. Some of these cars _20__by the company to the United States and Britain. In 1920, **Fiat___21___ making cars at a new** factory at Lingotto, near Turin. There was a track on the roof where the cars____22____ by technicians. In 1936, Fiat launched the Fiat 500. This car ___23___the Topolino – the Italian name for Mickey Mouse. The company grew, and in 1963 Fiat ___24___more than 300,000 vehicles. Today, Fiat is based in Turin, and its cars ___25___ all over the world.

19. A. produced B. was produced
20. A. exported B. were exported
21. A. started B. was started
22. A. tested B. were tested
23. A. called B. was called
24. A. exported B. was exported
25. A. sold B. are sold

Give one example each of the following (Q 26-30):

26. Split infinitive
27. A passive sentence beginning with 'Let'
28. 'Up' as an adverb
29. Linking verb
30. Anaphoric 'to'

[SET A]

Printed Pages: 8 **University Roll No...**

First Mid-Term Examination, Odd Semester 2019-20

B. Tech (CEA), II Year, Semester III

Ethics & Values: BELH-0006

Time: 1 Hour **Maximum Marks: 15**

Note: Read the passages carefully and answer the given questions. Each question is followed by multiple statements numbered 1,2,3,4, 5 & 6. Please note that there is only ONE CORRECT ANSWER (one among a, b, c, d) Choose the most appropriate one. **Please don't forget to mark the SET. on your OMR Sheet. (Attempt All Questions)**

Passage: 1 (1X6=6)

Imagineering is an art to explore an engineer within oneself. This is the very first ethics in the sense of one's commitment to rationality to observe, figure out and understand what really the case is. The duty is to develop engineering rationality in order to develop the functional ability what we call engineering imagination. No one would have thought in the past few decades how the world would change drastically in the way we experience our life now a days in a way almost technically equipped and digitally governed through software applications, gadgets, devices and systems. Software applications are changing the very means of communications, human psychology and lifestyles. Engineering imagination works through understanding a feature: it's process and design. Further, it provides the framework to form a new imagination for the present and future needs. The good Imagineering and the good engineering personality both require not only the technical skills but also the managerial strategies to develop the new software, device or product. whereas the technology impacts user psychology, while user satisfaction becomes the crucial fact for Imagineering or a new engineering design. Market acceptability demands easy and resistance free technology that

can be reached easily to all the users. Ethical handling based on triple P model argues that a technological creation must be people friendly and environment friendly then it should try to enhance the profitability. Gaming has involved all the generations of this time in a way that it has occupied a big space in human mind and hence affecting the psychology as we have witnessed how the Blue Whale game has taken many lives. It's not that all gaming software are bad. Some of them have designed with a thoughtful, creative and learning purpose so to provide fun and learning for the users. However, ethical considerations should be taken care of while working on Imagineering.

Questions:

1.Technical skills in Imagineering involve:

1. Market acceptability
2. Design
3. Process
4. User Satisfaction
5. Profitability
6. Features

a. 4,5, 6 b. 1, 2 ,3 ,4 c. 2, 3, 6 d. 3, 4 ,5, 6

2. Technology has changed the life as it is now:

1. Digitally governed
2. Less equipped
3. changing the means of communications
4. changed the lifestyles
5. Changed the psychology
6. less personal

a. 2,3,4,5 b. 1,3,4,5 c. only 1& 5 d. only 2 & 6

3. Ethical handling does not mean:
1. Triple P Model
2. Only People friendly approach
3. Only Environment friendly
4. more profit friendly
5. more people friendly and less environment friendly
6. All above
a. 1,2,3 b. 1,4,5 c. 1,2,34 d. 2,3,4,5
4. Facebook is an example of software Imagineering that exhibits:
1. engineering imagination
2. user satisfaction
3. means of communication
4. features and needs
5. problems in thinking
6. none of these
a. 1,2,4 b. 3,4,5 c. 2,3,4 d. 3,4,5,6
5. Managerial handling includes:
1. User satisfaction
2. Present and future needs
3. People and environment friendly approach
4. Profitability
5. Market acceptability
6. User psychology
a. only 1,2,3 b. only 4,5,6 c. 1,2,3,4,5,6 d. only 2,3,4
6. Ethical considerations should be taken care of in
1. Imagineering
2. User satisfaction
3. Software designs
4. Hardware and devices
5. Digital Designs
6. Triple P Model
a. Only 4,5,6 b. 1,2,6 c. only 4,5 d. only 3,4,5

Passage:2 (1X4=4)

The worth of studying ethics in engineering is not only in that that it gives an exposure to the understanding of professional ethics but also helps to understand one's personality in a deeper sense. Although without ethical applications any professional practice cannot conceived, it is must for the realization of one's abilities, potential talents, psychological solidarity, attitude and personality. Self-exploration is about observing, analyzing, figuring-out and understanding one's morality followed by the rationality. One of the dilemmas that we encounter at the outset is about one's own identity that lacks the integrity and affects our whole personality in a way we simply struggle to actually what we think, speak and live is not uniform, hence, it poses problems relating to truth, trust and temperament. Morality which is one's personal, subjective and individual opinions on what is right/wrong, good/bad, and ought/ought not. However, the capacity of moral judgment which is functioned by the rationality differs from that of our emotions, feelings, personal desires and mere inclinations. Moral judgment is based on a value judgment capacity that further follows a process of critical thinking, problematization-problem-solving and then decision making.

Questions:

7.The importance of ethics:

1. is only for professional development.
2. is only for personal development.
3. is for both 1&2
4. is only for society.
5. is only for environmental protection.
6. is for both 4&5

a. 1,2,4 b. 1,4,5 c. 1,2,4,5 d.3&6

8. What is problematization?
1. it's a form of Imagineering.
2. it involves creativity.
3. it only involves innovation.
4. it involves engineering rationality.
5. it requires analytical thinking
6. it is one's emotional tackling.

a. 1,2,4,5 b. 4,5,6 c. 2,4,5,6 d. 1,2,3

9. Self-exploration follows:
1. Rationality
2. Critical thinking
3. Problematization
4. Problem-solving
5. Moral-judgment
6. Value judgment

a. only 4,5,6 b. only 1,2,3 c. only 1,2,3,4 d. 1,2,3,4,5 &6

10. Morality is:
1. Personal
2. Subjective
3. Individualistic
4. based on only feelings and emotions.
5. based on reasons other than just desires.
6. One's moral opinion.

a. 1,2,3,4,5,6 b. 3,4,5,6 c. 1,2,3,5,6 d. only 2,3,4

Following questions are based on Case Study: The Violation of Privacy.
Questions: (.5X5=2.5)

11.What were the ethical concerns in the case?

1. Privacy of the clients.
2. Privacy of the doctors.
3. User's information.
4. Company's integrity
5. Violation of privacy.

a. 1,2,5 b. 1,3,4,5 c.2,3,4 d.1,2,3,4,5

12. What was **wrong** about Marcus's professional ethics?

1. Supplying the information
2. Protecting the information
3. Working for company's interest.
4. Minding his own interests.
5. Minding the user's privacy.

a. 1,2,3,5 b. 3,4,5 c. 2,4,5 d. 2,3,4

13. What was Marcus's accountability?

1. Work for the marketing department.
2. Work for the software application's technical issues.
3. Work for his own development only.
4. Work for the company's profit only.
5. Work for the user's interest only.

a. 1,2 ,3 ,4 b. 2,3 ,4, 5 c. Only 2 d. 1,2,3,4,5

14. What did you learn about the Professional Ethics from the case?

1. Following the right codes of software engineering.
2. Maintaining Privacy.
3. We should violate ethics for personal interest.
4. Software engineer should mind his/her growth only.
5. Privacy cannot be violated for personal interests.

1. 1,2,5 b. Only 1&2 c. Only 2 & 5 d. 3,4,5

15. What could be the ethical use of information in the case?

1. Leaking the user's identity.
2. information for company's profit.
3. information for Marcus's own growth.
4. Information for doctor's benefit.
5. None of the above.

a. Only 1,2,3 b. only 2&3 c. Only 5 d. Only 3&5

Following questions are based on video sessions(.5X5=2.5)

16. The girl thrown the mask in the movie: IDENTITY

1. She came to know about the reality.
2. She found everyone in integrity.
3. She better realized her integrity.
4. She tried to be original.
5. Mask was broken.

a. 1,3,4 b. 3,4,5 c. only 2&5 d.1,2,3

17. What is an ethical insight from the movie scene CORPORATE:

1. Ethics in profession really matters.
2. Professional competition must value people first.
3. Profit is the only bottom line.
4. Resignation of the person was an ethical response.
5. He should not have resigned.

a.3,4,5 b. 1,2,4 c. only 1,2,3 d. 1,2,3,4,5

18. What is an ethical insight from the Poetry scene SAHI AUR GALAT:

1. Personal morality should be respected.
2. Personal freedom should be respected.
3. Society is biased to a great extent.
4. Being morally emotional is not about moral judgment.
5. Moral judgment requires moral rationality.

a. only 1,2,3 b. only 3,4,5 c. 1,2,3,4,5 d. only 4&5

19. What was the ethical lesson from the first scene of the movie RIGHT HERE RIGHT NOW:

1. Anger leads to happiness
2. Behavior works in a chain reaction.
3. Anger leads to untoward events.
4. Good motivation and positive support leads to happiness
5. Good behavior and positive vibes saves from unfortunate events.

a. only 1,2,3 b. only 1,2,3,4 c. only 4&5 d.2,3,4,5

20. Which is **not** the correct ethical conclusion from the last scene of the Movie:

1. The man betrayed the beloved.
2. The man behaved in an unprofessional manner.
3. The urgency was to save the person.
4. The urgency was to go with the beloved.
5. Breaking the promise man responded unethically.

a. 1,2,4,5 b. 3,4,5 c. 1,3,4,5 d.1,2,3,4,5

[SET B]

Printed Pages: 8 University Roll No...

First Mid-Term Examination, Odd Semester 2019-20

B. Tech (CEA), II Year, Semester III

Ethics & Values: BELH-0006

Time: 1 Hour Maximum Marks: 15

Note: Read the passages carefully and answer the given questions. Each question is followed by multiple statements numbered 1,2,3,4, 5 & 6. Please note that there is only ONE CORRECT ANSWER (one among a, b, c, d) Choose the most appropriate one, **Please don't forget to mark the SET. on your OMR Sheet. (Attempt All Questions)**

Passage: 1 (1X4=4)

The worth of studying ethics in engineering is not only in that that it gives an exposure to the understanding of professional ethics but also helps to understand one's personality in a deeper sense. Although without ethical applications any professional practice cannot conceived, it is must for the realization of one's abilities, potential talents, psychological solidarity, attitude and personality. Self-exploration is about observing, analyzing, figuring-out and understanding one's morality followed by the rationality. One of the dilemmas that we encounter at the outset is about one's own identity that lacks the integrity and affects our whole personality in a way we simply struggle to actually what we think, speak and live is not uniform, hence, it poses problems relating to truth, trust and temperament. Morality which is one's personal, subjective and individual opinions on what is right/wrong, good/bad, and ought/ought not. However, the capacity of moral judgment which is functioned by the rationality differs from that of our emotions, feelings, personal desires and mere inclinations. Moral judgment is based on a value judgment capacity that further follows a

process of critical thinking, problematization-problem-solving and then decision making.

Questions:

1.The importance of ethics:

1. is only for professional development.
2. is only for personal development.
3. is for both 1&2
4. is only for society.
5. is only for environmental protection.
6. is for both 4&5

a. 1,2,4 b. 1,4,5 c. 1,2,4,5 d.3&6

2. What is problematization?

1. it's a form of Imagineering.
2. it involves creativity.
3. it only involves innovation.
4. it involves engineering rationality.
5. it requires analytical thinking
6. it is one's emotional tackling.

a. 1,2,4,5 b. 4,5,6 c. 2,4,5,6 d. 1,2,3

3. Self-exploration follows:

1. Rationality
2. Critical thinking
3. Problematization
4. Problem-solving
5. Moral-judgment
6. Value judgment

a. only 4,5,6 b. only 1,2,3 c. only 1,2,3,4 d. 1,2,3,4,5 &6

4. Morality is:
1. Personal
2. Subjective
3. Individualistic
4. based on only feelings and emotions.
5. based on reasons other than just desires.
6. One's moral opinion.

a. 1,2,3,4,5,6 b. 3,4,5,6 c. 1,2,3,5,6 d. only 2,3,4

Passage: 2 (1X6=6)

Imagineering is an art to explore an engineer within oneself. This is the very first ethics in the sense of one's commitment to rationality to observe, figure out and understand what really the case is. The duty is to develop engineering rationality in order to develop the functional ability what we call engineering imagination. No one would have thought in the past few decades how the world would change drastically in the way we experience our life now a days in a way almost technically equipped and digitally governed through software applications, gadgets, devices and systems. Software applications are changing the very means of communications, human psychology and lifestyles. Engineering imagination works through understanding a feature: it's process and design. Further, it provides the framework to form a new imagination for the present and future needs. The good Imagineering and the good engineering personality both require not only the technical skills but also the managerial strategies to develop the new software, device or product. whereas the technology impacts user psychology, while user satisfaction becomes the crucial fact for Imagineering or a new engineering design. Market acceptability demands easy and resistance free technology that can be reached easily to all the users. Ethical handling based on triple P model argues that a technological creation must be people friendly and environment friendly then it should try to enhance the profitability.

Gaming has involved all the generations of this time in a way that it has occupied a big space in human mind and hence affecting the psychology as we have witnessed how the Blue Whale game has taken many lives. It's not that all gaming software are bad. Some of them have designed with a thoughtful, creative and learning purpose so to provide fun and learning for the users. However, ethical considerations should be taken care of while working on Imagineering.

Questions:

5.Technical skills in Imagineering involve:

1. Market acceptability
2. Design
3. Process
4. User Satisfaction
5. Profitability
6. Features

a. 4,5, 6 b. 1, 2 ,3 ,4 c. 2, 3, 6 d. 3, 4 ,5, 6

6. Technology has changed the life as it is now:

1. Digitally governed
2. Less equipped
3. changing the means of communications
4. changed the lifestyles
5. Changed the psychology
6. less personal

a. 2,3,4,5 b. 1,3,4,5 c. only 1& 5 d. only 2 & 6

7. Ethical handling does not mean:

1. Triple P Model
2. Only People friendly approach
3. Only Environment friendly
4. more profit friendly
5. more people friendly and less environment friendly
6. All above

a. 1,2,3 b. 1,4,5 c. 1,2,34 d. 2,3,4,5

8. Facebook is an example of software Imagineering that exhibits:
1. engineering imagination
2. user satisfaction
3. means of communication
4. features and needs
5. problems in thinking
6. none of these
a. 1,2,4 b. 3,4,5 c. 2,3,4 d. 3,4,5,6
9. Managerial handling includes:
1. User satisfaction
2. Present and future needs
3. People and environment friendly approach
4. Profitability
5. Market acceptability
10. User psychology
a. only 1,2,3 b. only 4,5,6 c. 1,2,3,4,5,6 d. only 2,3,4
6. Ethical considerations should be taken care of in
1. Imagineering
2. User satisfaction
3. Software designs
4. Hardware and devices
5. Digital Designs
6. Triple P Model
a. Only 4,5,6 b. 1,2,6 c. only 4,5 d. only 3,4,5

Following questions are based on Case Study: The Violation of Privacy.

Questions: (.5X5=2.5)

11.What were the ethical concerns in the case?

1. Privacy of the clients.
2. Privacy of the doctors.
3. User's information.
4. Company's integrity
5. Violation of privacy.

a. 1,2,5 b. 1,3,4,5 c.2,3,4 d.1,2,3,4,5

12. What was **wrong** about Marcus's professional ethics?

1. Supplying the information
2. Protecting the information
3. Working for company's interest.
4. Minding his own interests.
5. Minding the user's privacy.

a. 1,2,3,5 b. 3,4,5 c. 2,4,5 d. 2,3,4

13. What was Marcus's accountability?

1. Work for the marketing department.
2. Work for the software application's technical issues.
3. Work for his own development only.
4. Work for the company's profit only.
5. Work for the user's interest only.

a. 1,2 ,3 ,4 b. 2,3 ,4, 5 c. Only 2 d. 1,2,3,4,5

14. What did you learn about the Professional Ethics from the case?

1. Following the right codes of software engineering.
2. Maintaining Privacy.
3. We should violate ethics for personal interest.
4. Software engineer should mind his/her growth only.
5. Privacy cannot be violated for personal interests.

1. 1,2,5 b. Only 1&2 c. Only 2 & 5 d. 3,4,5

15. What could be the ethical use of information in the case?

1. Leaking the user's identity.
2. information for company's profit.
3. information for Marcus's own growth.
4. Information for doctor's benefit.
5. None of the above.

a. Only 1,2,3 b. only 2&3 c. Only 5 d. Only 3&5

Following questions are based on video sessions: (.5X5=2.5)

16. The girl thrown the mask in the movie: IDENTITY

1. She came to know about the reality.
2. She found everyone in integrity.
3. She better realized her integrity.
4. She tried to be original.
5. Mask was broken.

a. 1,3,4 b. 3,4,5 c. only 2&5 d.1,2,3

17. What is an ethical insight from the movie scene CORPORATE:

1. Ethics in profession really matters.
2. Professional competition must value people first.
3. Profit is the only bottom line.
4. Resignation of the person was an ethical response.
5. He should not have resigned.

a.3,4,5 b. 1,2,4 c. only 1,2,3 d. 1,2,3,4,5

18. What is an ethical insight from the Poetry scene SAHI AUR GALAT:

1. Personal morality should be respected.
2. Personal freedom should be respected.
3. Society is biased to a great extent.
4. Being morally emotional is not about moral judgment.
5. Moral judgment requires moral rationality.

a. only 1,2,3 b. only 3,4,5 c. 1,2,3,4,5 d. only 4&5

19. What was the ethical lesson from the first scene of the movie RIGHT HERE RIGHT NOW:

1. Anger leads to happiness
2. Behavior works in a chain reaction.
3. Anger leads to untoward events.
4. Good motivation and positive support leads to happiness
5. Good behavior and positive vibes saves from unfortunate events.

a. only 1,2,3 b. only 1,2,3,4 c. only 4&5 d.2,3,4,5

20. Which is **not** the correct ethical conclusion from the last scene of the Movie:

1. The man betrayed the beloved.
2. The man behaved in an unprofessional manner.
3. The urgency was to save the person.
4. The urgency was to go with the beloved.
5. Breaking the promise man responded unethically.

a. 1,2,4,5 b. 3,4,5 c. 1,3,4,5 d.1,2,3,4,5

[SET C]

Printed Pages:8 University Roll No...

First Mid-Term Examination, Odd Semester 2019-20

B. Tech (CEA), II Year, Semester III

Ethics & Values: BELH-0006

Time: 1 Hour Maximum Marks: 15

Note: Read the passages carefully and answer the given questions. Each question is followed by multiple statements numbered 1,2,3,4, 5 & 6. Please note that there is only ONE CORRECT ANSWER (one among a, b, c, d) Choose the most appropriate one. **Please don't forget to mark the SET. on your OMR Sheet. (Attempt All Questions)**

Passage: 1 (1X6=6)

Imagineering is an art to explore an engineer within oneself. This is the very first ethics in the sense of one's commitment to rationality to observe, figure out and understand what really the case is. The duty is to develop engineering rationality in order to develop the functional ability what we call engineering imagination. No one would have thought in the past few decades how the world would change drastically in the way we experience our life now a days in a way almost technically equipped and digitally governed through software applications, gadgets, devices and systems. Software applications are changing the very means of communications, human psychology and lifestyles. Engineering imagination works through understanding a feature: it's process and design. Further, it provides the framework to form a new imagination for the present and future needs. The good Imagineering and the good engineering personality both require not only the technical skills but also the managerial strategies to develop the new software, device or product. whereas the technology impacts user psychology, while user satisfaction becomes the crucial fact for Imagineering or a new engineering design. Market acceptability demands easy and resistance free technology that

can be reached easily to all the users. Ethical handling based on triple P model argues that a technological creation must be people friendly and environment friendly then it should try to enhance the profitability. Gaming has involved all the generations of this time in a way that it has occupied a big space in human mind and hence affecting the psychology as we have witnessed how the Blue Whale game has taken many lives. It's not that all gaming software are bad. Some of them have designed with a thoughtful, creative and learning purpose so to provide fun and learning for the users. However, ethical considerations should be taken care of while working on Imagineering.

Questions:

1.Technical skills in Imagineering involve:

1. Market acceptability
2. Design
3. Process
4. User Satisfaction
5. Profitability
6. Features

a. 4,5, 6 b. 1, 2 ,3 ,4 c. 2, 3, 6 d. 3, 4 ,5, 6

2. Technology has changed the life as it is now:

1. Digitally governed
2. Less equipped
3. changing the means of communications
4. changed the lifestyles
5. Changed the psychology
6. less personal

a. 2,3,4,5 b. 1,3,4,5 c. only 1& 5 d. only 2 & 6

3. Ethical handling does not mean:
1. Triple P Model
2. Only People friendly approach
3. Only Environment friendly
4. more profit friendly
5. more people friendly and less environment friendly
6. All above

a. 1,2,3 b. 1,4,5 c. 1,2,34 d. 2,3,4,5

4. Facebook is an example of software Imagineering that exhibits:
1. engineering imagination
2. user satisfaction
3. means of communication
4. features and needs
5. problems in thinking
6. none of these

a. 1,2,4 b. 3,4,5 c. 2,3,4 d. 3,4,5,6

5. Managerial handling includes:
1. User satisfaction
2. Present and future needs
3. People and environment friendly approach
4. Profitability
5. Market acceptability
6. User psychology

a. only 1,2,3 b. only 4,5,6 c. 1,2,3,4,5,6 d. only 2,3,4

6. Ethical considerations should be taken care of in
1. Imagineering
2. User satisfaction
3. Software designs
4. Hardware and devices
5. Digital Designs
6. Triple P Model

a. Only 4,5,6 b. 1,2,6 c. only 4,5 d. only 3,4,5

Passage:2 (1X4=4)

The worth of studying ethics in engineering is not only in that that it gives an exposure to the understanding of professional ethics but also helps to understand one's personality in a deeper sense. Although without ethical applications any professional practice cannot conceived, it is must for the realization of one's abilities, potential talents, psychological solidarity, attitude and personality. Self-exploration is about observing, analyzing, figuring-out and understanding one's morality followed by the rationality. One of the dilemmas that we encounter at the outset is about one's own identity that lacks the integrity and affects our whole personality in a way we simply struggle to actually what we think, speak and live is not uniform, hence, it poses problems relating to truth, trust and temperament. Morality which is one's personal, subjective and individual opinions on what is right/wrong, good/bad, and ought/ought not. However, the capacity of moral judgment which is functioned by the rationality differs from that of our emotions, feelings, personal desires and mere inclinations. Moral *judgment is based on a value judgment capacity that further follows a* process of critical thinking, problematization-problem-solving and then decision making.

Questions:

7.The importance of ethics:

1. is only for professional development.
2. is only for personal development.
3. is for both 1&2
4. is only for society.
5. is only for environmental protection.
6. is for both 4&5

a. 1,2,4 b. 1,4,5 c. 1,2,4,5 d.3&6

8. What is problematization?
1. it's a form of Imagineering.
2. it involves creativity.
3. it only involves innovation.
4. it involves engineering rationality.
5. it requires analytical thinking
6. it is one's emotional tackling.

a. 1,2,4,5 b. 4,5,6 c. 2,4,5,6 d. 1,2,3

9. Self-exploration follows:
1. Rationality
2. Critical thinking
3. Problematization
4. Problem-solving
5. Moral-judgment
6. Value judgment

a. only 4,5,6 b. only 1,2,3 c. only 1,2,3,4 d. 1,2,3,4,5 &6

10. Morality is:
1. Personal
2. Subjective
3. Individualistic
4. based on only feelings and emotions.
5. based on reasons other than just desires.
6. One's moral opinion.

a. 1,2,3,4,5,6 b. 3,4,5,6 c. 1,2,3,5,6 d. only 2,3,4

Following questions are based on video sessions(.5X5=2.5)

11. The girl thrown the mask in the movie: IDENTITY

1. She came to know about the reality.
2. She found everyone in integrity.
3. She better realized her integrity.
4. She tried to be original.
5. Mask was broken.

a. 1,3,4 b. 3,4,5 c. only 2&5 d.1,2,3

12. What is an ethical insight from the movie scene CORPORATE:

1. Ethics in profession really matters.
2. Professional competition must value people first.
3. Profit is the only bottom line.
4. Resignation of the person was an ethical response.
5. He should not have resigned.

a.3,4,5 b. 1,2,4 c. only 1,2,3 d. 1,2,3,4,5

13. What is an ethical insight from the Poetry scene SAHI AUR GALAT:

1. Personal morality should be respected.
2. Personal freedom should be respected.
3. Society is biased to a great extent.
4. Being morally emotional is not about moral judgment.
5. Moral judgment requires moral rationality.

a. only 1,2,3 b. only 3,4,5 c. 1,2,3,4,5 d. only 4&5

14. What was the ethical lesson from the first scene of the movie RIGHT HERE RIGHT NOW:

1. Anger leads to happiness
2. Behavior works in a chain reaction.
3. Anger leads to untoward events.
4. Good motivation and positive support leads to happiness
5. Good behavior and positive vibes saves from unfortunate events.

a. only 1,2,3 b. only 1,2,3,4 c. only 4&5 d.2,3,4,5

15. Which is **not** the correct ethical conclusion from the last scene of the Movie:

1. The man betrayed the beloved.
2. The man behaved in an unprofessional manner.
3. The urgency was to save the person.
4. The urgency was to go with the beloved.
5. Breaking the promise man responded unethically.

a. 1,2,4,5 b. 3,4,5 c. 1,3,4,5 d.1,2,3,4,5

Following questions are based on Case Study: The Violation of Privacy.

Questions: (.5X5=2.5)

16.What were the ethical concerns in the case?

1. Privacy of the clients.
2. Privacy of the doctors.
3. User's information.
4. Company's integrity
5. Violation of privacy.

a. 1,2,5 b. 1,3,4,5 c.2,3,4 d.1,2,3,4,5

17. What was wrong about Marcus's professional ethics?

1. Supplying the information
2. Protecting the information
3. Working for company's interest.
4. Minding his own interests
5. Minding the user's privacy.

a. 1,2,3,5 b. 3,4,5 c. 2,4,5 d. 2,3,4

18. What was Marcus's accountability?

1. Work for the marketing department.
2. Work for the software application's technical issues.
3. Work for his own development only.
4. Work for the company's profit only.
5. Work for the user's interest only.

a. 1,2 ,3 ,4 b. 2,3 ,4, 5 c. Only 2 d. 1,2,3,4,5

19. What did you learn about the Professional Ethics from the case?

1. Following the right codes of software engineering.
2. Maintaining Privacy.
3. We should violate ethics for personal interest.
4. Software engineer should mind his/her growth only.
5. Privacy cannot be violated for personal interests.

1. 1,2,5 b. Only 1&2 c. Only 2 & 5 d. 3,4,5

20. What could be the ethical use of information in the case?

1. Leaking the user's identity.
2. information for company's profit.
3. information for Marcus's own growth.
4. Information for doctor's benefit.
5. None of the above.

a. Only 1,2,3 b. only 2&3 c. Only 5 d. Only 3&5

[SET D]

Printed Pages: 8 University Roll No...

First Mid-Term Examination, Odd Semester 2019-20

B. Tech (CEA), II Year, Semester III

Ethics & Values: BELH-0006

Time: 1 Hour Maximum Marks: 15

Note: Read the passages carefully and answer the given questions. Each question is followed by multiple statements numbered 1,2,3,4, 5 & 6. Please note that there is only ONE CORRECT ANSWER (one among a, b, c, d) Choose the most appropriate one. **Please don't forget to mark the SET. on your OMR Sheet. (Attempt All Questions)**

Passage: 1 (1X6=6)

Imagineering is an art to explore an engineer within oneself. This is the very first ethics in the sense of one's commitment to rationality to observe, figure out and understand what really the case is. The duty is to develop engineering rationality in order to develop the functional ability what we call engineering imagination. No one would have thought in the past few decades how the world would change drastically in the way we experience our life now a days in a way almost technically equipped and digitally governed through software applications, gadgets, devices and systems. Software applications are changing the very means of communications, human psychology and lifestyles. Engineering imagination works through understanding a feature: it's process and design. Further, it provides the framework to form a new imagination for the present and future needs. The good Imagineering and the good engineering personality both require not only the technical skills but also the managerial strategies to develop the new software, device or product. whereas the technology impacts user psychology, while user satisfaction becomes the crucial fact for Imagineering or a new engineering design. Market acceptability demands easy and resistance free technology that

can be reached easily to all the users. Ethical handling based on triple P model argues that a technological creation must be people friendly and environment friendly then it should try to enhance the profitability. Gaming has involved all the generations of this time in a way that it has occupied a big space in human mind and hence affecting the psychology as we have witnessed how the Blue Whale game has taken many lives. It's not that all gaming software are bad. Some of them have designed with a thoughtful, creative and learning purpose so to provide fun and learning for the users. However, ethical considerations should be taken care of while working on Imagineering.

Questions:

1.Technical skills in Imagineering involve:

1. Market acceptability
2. Design
3. Process
4. User Satisfaction
5. Profitability
6. Features

a. 4,5, 6 b. 1, 2 ,3 ,4 c. 2, 3, 6 d. 3, 4 ,5, 6

2. Technology has changed the life as it is now:

1. Digitally governed
2. Less equipped
3. changing the means of communications
4. changed the lifestyles
5. Changed the psychology
6. less personal

a. 2,3,4,5 b. 1,3,4,5 c. only 1& 5 d. only 2 & 6

3. Ethical handling does not mean:
1. Triple P Model
2. Only People friendly approach
3. Only Environment friendly
4. more profit friendly
5. more people friendly and less environment friendly
6. All above
a. 1,2,3 b. 1,4,5 c. 1,2,34 d. 2,3,4,5
4. Facebook is an example of software Imagineering that exhibits:
1. engineering imagination
2. user satisfaction
3. means of communication
4. features and needs
5. problems in thinking
6. none of these
a. 1,2,4 b. 3,4,5 c. 2,3,4 d. 3,4,5,6
5. Managerial handling includes:
1. User satisfaction
2. Present and future needs
3. People and environment friendly approach
4. Profitability
5. Market acceptability
6. User psychology
a. only 1,2,3 b. only 4,5,6 c. 1,2,3,4,5,6 d. only 2,3,4
6. Ethical considerations should be taken care of in
1. Imagineering
2. User satisfaction
3. Software designs
4. Hardware and devices
5. Digital Designs
6. Triple P Model
a. Only 4,5,6 b. 1,2,6 c. only 4,5 d. only 3,4,5

Following questions are based on Case Study: The Violation of Privacy.

Questions: (.5X5=2.5)

7.What were the ethical concerns in the case?

1. Privacy of the clients.
2. Privacy of the doctors.
3. User's information.
4. Company's integrity
5. Violation of privacy.

a. 1,2,5 b. 1,3,4,5 c.2,3,4 d.1,2,3,4,5

8. What was **wrong** about Marcus's professional ethics?

1. Supplying the information
2. Protecting the information
3. Working for company's interest.
4. Minding his own interests.
5. Minding the user's privacy.

a. 1,2,3,5 b. 3,4,5 c. 2,4,5 d. 2,3,4

9. What was Marcus's accountability?

1. Work for the marketing department.
2. Work for the software application's technical issues.
3. Work for his own development only.
4. Work for the company's profit only.
5. Work for the user's interest only.

a. 1,2 ,3 ,4 b. 2,3 ,4, 5 c. Only 2 d. 1,2,3,4,5

10. What did you learn about the Professional Ethics from the case?

1. Following the right codes of software engineering.
2. Maintaining Privacy.
3. We should violate ethics for personal interest.
4. Software engineer should mind his/her growth only.
5. Privacy cannot be violated for personal interests.

1. 1,2,5 b. Only 1&2 c. Only 2 & 5 d. 3,4,5

11. What could be the ethical use of information in the case?

1. Leaking the user's identity.
2. information for company's profit.
3. information for Marcus's own growth.
4. Information for doctor's benefit.
5. None of the above.

a. Only 1,2,3 b. only 2&3 c. Only 5 d. Only 3&5

Following questions are based on video sessions:(.5X5=2.5)

12. The girl thrown the mask in the movie: IDENTITY

1. She came to know about the reality.
2. She found everyone in integrity.
3. She better realized her integrity.
4. She tried to be original.
5. Mask was broken.

a. 1,3,4 b. 3,4,5 c. only 2&5 d.1,2,3

13. What is an ethical insight from the movie scene CORPORATE:

1. Ethics in profession really matters.
2. Professional competition must value people first.
3. Profit is the only bottom line.
4. Resignation of the person was an ethical response.
5. He should not have resigned.

a.3,4,5 b. 1,2,4 c. only 1,2,3 d. 1,2,3,4,5

14. What is an ethical insight from the Poetry scene SAHI AUR GALAT:

1. Personal morality should be respected.
2. Personal freedom should be respected.
3. Society is biased to a great extent.
4. Being morally emotional is not about moral judgment.
5. Moral judgment requires moral rationality.

a. only 1,2,3 b. only 3,4,5 c. 1,2,3,4,5 d. only 4&5

15. What was the ethical lesson from the first scene of the movie RIGHT HERE RIGHT NOW:

1. Anger leads to happiness
2. Behavior works in a chain reaction.
3. Anger leads to untoward events.
4. Good motivation and positive support leads to happiness
5. Good behavior and positive vibes saves from unfortunate events.

a. only 1,2,3 b. only 1,2,3,4 c. only 4&5 d.2,3,4,5

16. Which is **not** the correct ethical conclusion from the last scene of the Movie:

1. The man betrayed the beloved.
2. The man behaved in an unprofessional manner.
3. The urgency was to save the person.
4. The urgency was to go with the beloved.
5. Breaking the promise man responded unethically.

a. 1,2,4,5 b. 3,4,5 c. 1,3,4,5 d.1,2,3,4,5

Passage:2 (1X4=4)

The worth of studying ethics in engineering is not only in that that it gives an exposure to the understanding of professional ethics but also helps to understand one's personality in a deeper sense. Although without ethical applications any professional practice cannot conceived, it is must for the realization of one's abilities, potential talents, psychological solidarity, attitude and personality. Self-exploration is about observing, analyzing, figuring-out and understanding one's morality followed by the rationality. One of the dilemmas that we encounter at the outset is about one's own identity that lacks the integrity and affects our whole personality in a way we simply struggle to actually what we think, speak and live is not uniform, hence, it poses problems relating to truth, trust and temperament. Morality which is one's personal, subjective and individual opinions on what is

right/wrong, good/bad, and ought/ought not. However, the capacity of moral judgment which is functioned by the rationality differs from that of our emotions, feelings, personal desires and mere inclinations. Moral judgment is based on a value judgment capacity that further follows a process of critical thinking, problematization-problem-solving and then decision making.

Questions:

17.The importance of ethics:

1. is only for professional development.
2. is only for personal development.
3. is for both 1&2
4. is only for society.
5. is only for environmental protection.
6. is for both 4&5

a. 1,2,4 b. 1,4,5 c. 1,2,4,5 d.3&6

18. What is problematization?
1. it's a form of Imagineering.
2. it involves creativity.
3. it only involves innovation.
4. it involves engineering rationality.
5. it requires analytical thinking
6. it is one's emotional tackling.
a. 1,2,4,5 b. 4,5,6 c. 2,4,5,6 d. 1,2,3

19. Self-exploration follows:
1. Rationality
2. Critical thinking
3. Problematization
4. Problem-solving
5. Moral-judgment
6. Value judgment
a. only 4,5,6 b. only 1,2,3 c. only 1,2,3,4 d. 1,2,3,4,5 &6

20. Morality is:
1. Personal
2. Subjective
3. Individualistic
4. based on only feelings and emotions.
5. based on reasons other than just desires.
6. One's moral opinion.
a. 1,2,3,4,5,6 b. 3,4,5,6 c. 1,2,3,5,6 d. only 2,3,4

University Roll No.....

First Mid-Term Examination, Odd Semester 2019-20

B.Tech (CSE), 2nd year, 3rd Semester

Subject Code: Database Management System (BCSC0003)

Time: 1 Hour **Maximum Marks: 15**

Section – A

Attempt all three question **3 x 2 = 6 Marks**

1. What is the importance of data independence in the context of DBMS? Explain with suitable examples.
2. Consider the relations r(A, B) and s(B, C), where s.B is a primary key and r.B is a foreign key referencing s.B. Consider the query Q: $r \bowtie (\sigma B<5(s))$
   Let LOJ denote the natural left outer-join operation. Assume that r and s contain no null values. Which one of the following is NOT equivalent to Q, justify?
   a) $\sigma_{B<5}(r \text{ LOJ } s)$
   b) $r \text{ LOJ } (\sigma_{B<5}(s))$
3. Discuss how the database approach differs from traditional file systems.

Section – B

Attempt all three question **3 x 3 = 9 Marks**

1. Consider the following requirement set and draw ER diagram. also reduced ER diagram into table.
   - The company is organized into DEPARTMENTs. Each department has a name, number and an employee who *manages* the department. We keep track of the start date of the department manager.
   - Each department *controls* a number of PROJECTs. Each project has a name, number and is located at a single location.

- We consider the employee having three categories staff, technical staff, engineer. And some time technical staff perform as an engineer. We store each EMPLOYEE's social security number, address, salary, sex, birthdate, and more than one degree. Each employee *works for* one department and some of the employee *work on* several projects. We keep track of the number of hours per week that an employee currently works on each project.
- Each employee may *have* a number of DEPENDENTs. For each dependent, we keep track of their name, sex, birthdate, and relationship to employee.

2. What are different type of end users explain each in brief? And also explain the role of DBA?
3. Using these relations, give answers to following questions in Relational Algebra:

employee (person-name, street, city)

works (person-name, company-name, salary)

company (company-name, city)

manages (person-name, manager-name)

a) Find the names of all employees who work for First Bank Corporation.

b) Find the names and cities of residence of all employees who work for First Bank Corporation.

c) Find the names of all employees in this database who do not work for First Bank Corporation.

d) Find the names of all employees who earn more than every employee of Small Bank Corporation.

e) Assume the companies may be located in several cities. Find all companies located in every city in which Small Bank Corporation is located.

f) Using above relations, give answers the questions in Relational calculus: Display the record of those employee who lived in 'Mathura'.

Paper Code-BTDH 0301	SET-C

Soft Skills

First Mid Term, Odd Sem Examination (2019-20)

Course- B.tech **Year- II**

Branch- (All)

Time : 60 Min. **M.M-20**

Instructions :

1. **Attempt all questions.**
2. **There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.**
3. **The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.**
4. **The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.**
5. **Only black& blue pen should be used to fill forms.**
6. **Each branch is identified with a specific subject code which should be written in subject column of OMR sheet**
7. **Write your University roll number in the roll no column of OMR sheet.**

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:..... **All The Best**

(Aptitude)-No of question-40, marks-0.50

Q1. The number 94210p31q is divisible by 44. What is the value of p + q?
A.1 B. 5 C.6 D.8 E. None of these

Q2. Find the value of x if the number 58215x237 is divisible by 11?
A.1 B. 5 C.6 D.7 E. None of these

Q3. If the number 63576x2 is divisible by 8, find the least value that x can take.
A.1 B. 3 C.6 D.8 E. None of these

Q4. If the number 86y25x6 is divisible by 36, find the possible no of solutons.
A.1 B. 5 C.6 D.8 E. None of these

Q5. If the number 42573x is divisible by 72, find the value that x can assume.
A.1 B. 5 C.6 D.8 E. None of these

Q6. How many different values can x take if the number 2506x8 is divisible by 8?
A.1 B. 5 C.6 D.8 E. None of these

Q7. On what dates of January, 2001 did Sunday fall?
A. 1^{st}, 8^{th}, 15^{th}, 22^{nd}, 29^{th} B. 2^{nd}, 9^{th}, 16^{th}, 23^{rd}, 30^{th}
C. 3^{rd}, 10^{th}, 17^{th}, 24^{th} D. 7^{th}, 14^{th}, 21^{th}, 28^{th} E. None of these

Q8. The last day of a century cannot be
A.Monday B.Wednesday C.Thursday D.Friday
E. None of these

Q9. The calendar for the year 1897 will be the same for the year:
A.1909 B. 1903 C.1908 D.1907 E. None of these

Q10. On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec, 2006?
A.Sunday B.Thursday C.Tuesday D.Friday
E. None of these

Q11. Which of the following is a leap year?
A.1900 B. 2016 C.2100 D.2018
E. None of these

Q12. The angle between the minute hand and the hour hand of a clock when the time is 4.20, is:
A.0° B.10° C.5° D.20° E. None of these

Q13. At what time between 8 and 9, the angle between the minute hand and the hour hand of a clock will be 60°:

A. $32\frac{8}{11}$ B. $53\frac{6}{11}$ C.30 D. Both A and B

E. None of these

Q14. How many times in a day, are the hands of a clock in straight line but opposite in direction between 6 AM to 5 PM in a day?

A.11 B.12 C.22 D.24 E. None of these

Q15. An accurate clock shows 8 o'clock in the morning. Through how may degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?

A. 144° B. 150° C. 168° D. 180° E. None of these

Q16. A watch which gains 5 seconds in 3 minutes was set right at 7 a.m. In the afternoon of the same day, when the watch indicated quarter past 4 o'clock, the true time is:

A. $59\frac{7}{12}$ min. past 3 B. 4 p.m. C. $58\frac{7}{11}$ min. past 3

D. $2\frac{3}{11}$ min past 4 E. None of these

Q17. In 16 minutes, minute hand gains over hour hand by:

A. 16 B. 80 C. 88 D. 96 E. None of these

Q18. If the two hands in a clock are 3 minutes division apart, then the angle between them is

A. 3^0 B. 18^0 C. 24^0 D. 60^0 E. None of these

Q19. A clock is set right at 3 AM, the clock loses 24 minutes in 24 hours. What will be the true time when the clock indicates 10:40 PM?

A. 10:30 PM B. 11 PM C. 10:50 PM D. 11:30 PM

E. None of these

Q20. John was born on Feb 29th of 2012 which happened to be a Wednesday. If he lives to be 101 years old, how many birthdays would he celebrate on a Wednesday?

A. 3 B. 4 C. 5 D. 1 E. None of these

Q21. What will be the day of the week 15th August, 2010?

A. Sunday B. Monday C. Tuesday D.Friday

E. None of these

Q22. Today is Monday. After 61 days, it will be:

A. Sunday B. Monday C. Tuesday D.Saturday E. None of these

Q23. Two students appeared at an examination. One of them secured 9 marks more than the other and his marks was 56% of the sum of their marks. The marks obtained by them are?
A. 42, 33 B. 39,30 C. 41,32 D. 43,34 E. None of these

Q24. The price of a car is Rs. 3,25,000. It was insured to 85% of its price. The car was damaged completely in an accident and the insurance company paid 90% of the insurance. What was the difference between the price of the car and the amount received?
A. 70,000 B. 82,150 C. 76,375 D. 34,000 E. None of these

Q25. The price of diesel increases by 50 %. Find by how much percent a truck owner must reduce his consumption in order to maintain the same budget? (Approx)
A. 11% B. 22% C.33% D. 44% E. None of these

Q26. Two friends, Akash & Beenu had some candies each. One of them had 15 candies more than the other. The candies with Akash was 60% of the total candies with them. How many candies did each have?
A. 40, 25 B. 47, 32 C. 45, 30 D. 49, 34 E. None of these

Q27. What percentage of numbers from 1 to 30 has 1 or 9 in the unit's digit?
A. 12 B. 15 C. 20 D. 22 E. None of these

Q28. In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:
A. 2700 B. 2900 C. 3000 D. 3100 E. None of these

Q29. The population of different trees in a field increased by 10 % in first year, increased by 8 % in second year and decreased by 10 % in third year. If at present the number of trees is 26730, then find the number of tress in the beginning.
A. 30000 B. 25000 C. 27000 D. 27865 E. None of these

Q30. A student attempts x number of questions. He answers 15 correctly out of first 20 questions and of the remaining questions, he answers 1/3 correctly. If all questions have same credit and the student gets 50 % marks, then find the value of x.
A. 30 B. 35 C. 45 D. 50 E. None of these

Q31. The current birth rate per thousand is 30, whereas corresponding death rate is 10 per thousand. Find the net growth rate in terms of population increase in percent.
A. 1.5 % B. 2 % C. 2.5 % D. 3 % E. None of these

Q32. When 35 is subtracted from a number; it reduces to its 80 %. Find the four-fifth of that number.
A. 140 B. 125 C. 137 D.129 E. None of these

Q33. $\sqrt[2]{144} + \sqrt[3]{1331} + + \sqrt[2]{125} =$.....
A. 47.18 B. 48.23 C. 46.34 D. 49.28 E. None of these

Q34. $45^2 + 54^2 + 48^2 + \sqrt[2]{1024} =$.....
A. 7287 B. 7277 C. 7239 D. 7284 E. None of these

Q35. $\sqrt[3]{2744} + \sqrt[2]{996004} + 63^2 =$.....
A. 3969 B. 4012 C. 4064 D. 3982 E. None of these

Q36. $65^2 + 64^2 + 63^2 + \sqrt[2]{125} + \sqrt[3]{36.24} =$.....
A. 12230.33 B. 12231.44 C. 12307.2 D. 12308.5
E. None of these

Q37. $101^2 + 102^2 + 103^2 + 105^2 + 106^2 =$.....
A. 53405 B. 53412 C. 54445 D. 53475 E. None of these

Q38. $\sqrt[3]{2200} + \sqrt[2]{48} + 42 \times 11 =$.....
A. 484.6 B. 481.9 C. 483.2 D. 485.4 E. None of these

Q39. 13% of 1100 + 17% of 2100 =? + 26% of 350
A. 409 B. 411 C. 413 D. 415 E. None of these

Q40. $1331 \times 121 \times 0.11 = (1.1)? \times 11000$
A. 5 B. 7 C. 4 D. 3 E. None of these

Printed Pages: 1 **University Roll No.....**

First Term Examination, Odd Semester 2019-20

Program: B.Tech.(CSE) **Year: II** **Semester: III**

Sub Code: BCSC0010 **Sub Name: Discrete Mathematics**

Time: 1 Hour **Maximum Marks: 15**

Section – A

Note: Attempt All Questions. *3 x 2 = 6 Marks*

Q.1 Prove that $X - (Y \cap Z) = (X - Y) \cup (X - Z)$ by principle of equality.

Q.2 Prove that the relation $R = \{(x, y) \mid x - y \text{ is an even integer for all } x, y \in Z\}$ is an equivalence relation.

Q.3 In an examination, 70% of the students passed in Hindi, 66% in English, 25% failed in both the subjects and 220 passed in both the subjects. Find the total number of students.

Section – B

Note: Attempt All Questions. *3 x 3 = 9 Marks*

Q.1 Solve the recurrence relation: $y_r - 6y_{r-1} + 9y_{r-2} = (r+1)3^r$

Q.2 (i) How many arrangements can be made by using letters of "HINDUSTAN"?

(ii) How many of these arrangements begin and end with vowels?

(iii) If 20 persons are selected for presenting a cultural programme, then find out minimum number of persons who shall be assigned same day of the week.

Q.3 Apply generating function technique to solve the recurrence relation:

$a_{n+2} - 4a_{n+1} + 3a_n = 0$, with initial conditions $a_0 = 2, a_1 = 4$

First Mid Term Examination, 2019-20
Course: B.Tech. (CSE), Year:II, Semester:II
Software Engineering (BCSC 0009)

Uni. Roll No: _____

Time: 1 Hrs. **Total Marks: 15**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section A

Note: Attempt All Questions **(3x2=6)**

I. Explain Bath tub curve for Hardware with the help of diagram.

II. Design use case diagram for a hospital's reception. Hospital Reception subsystem or module supports some of the many job duties of a hospital receptionist. Receptionist schedules patient's appointment and admission to the hospital, collects information from the patient by phone and/or upon patient's arrival to the hospital. For the patient that will stay in the hospital, she or he should have a bed allotted in a ward. Receptionists might also receive patient's payments, record them in a database and provide receipts, file insurance claims and medical reports.

III. Assume that you are working as a Project Manager in any software development company. You are supposed to develop a software estimated 450 KLOC. You have a choice of hiring from two pools of developers mentioned in below cases.

Case 1: Very highly capable (PCAP = 0.70) with very little experience in the programming language (LEXP =1.14) being used or

Case 2: Developers of low quality (PCAP = 1.17) but a lot of experience with the programming language (LEXP = 0.95).

Compare the effort & duration of development for both the cases.

Section B

Note: Attempt All Questions **(3x3=9)**

I. Consider Airline Reservation System where User can login, search Flights, check seats Availability, reserve a ticket, make payment and generate a Bill. Admin can login, change Flight schedule and ticket Fair. Draw DFD Level-0, Level-1, Level-2. (Note: Do not forget to document your assumptions, if you make any)

II. A project is assigned to a software industry by a client. Which model should be used for software development under which requirements are changing rapidly and required software in different release. Team members size must be 3 -8 members for a module with project leader and requirements are gathers in form of user stories. Define the process of this model by explaining all the major components with suitable diagram.

III. Identify the Average Staff size of a project (Using COCOMO Model) with the following domain characteristics:

a. No. of I/P = 45(Low)
b. No. of O/P = 62(High)
c. No. of user Inquiries = 36(Avg)
d. No. of files = 23(High)
e. No. of external interfaces = 12(Low)

Assume that all the complexity adjustment values are average. Assume that 14 algorithms have been counted. The software is going to be implemented in C language.

Printed Pages: 1 **University Roll No.....**

First Mid-Term Examination, Odd Semester 2019-20

Programme: B.Tech. Branch: CSE(CCV) Year: II nd Year

Subject with Code: Introduction to IT Infrastructure Landscape , BCSE0501

Time : 1 Hour **Maximum Marks: 15**

Section – A

Three questions of 02 marks each (with no internal choice). 3 x 2 = 6 Marks

I. What do you understand by FC-AL? List all the properties FC-AL topology persists.

II. What is bare metal hypervisor? Discuss the types of hypervisors, with diagram?

III. "Zoning is very important while designing a Virtual Private Cloud." Analyze the statement by explaining the need of Zoning. Also explain the types of zoning in brief.

Section – B

Three questions of 03 marks each (with no internal choice). 3 x 3 = 9 Marks

I. What is storage virtualization? Explain all its types in detail with its advantages and disadvantages.

II. "RAID is one of the best technologies to implement SAN." Explain the statement. Explain RAID 5 implementation with an example of your own?

III. "LDAP plays an important role in cloud computing." Analyze the statement and explain the models of LDAP referred.

Printed Pages: 1 **University Roll No.....**

First Term Examination, Odd Semester 2019-20

Program: B.Tech.CSE(CSF) Year: II, Semester: III

Sub Code: BCSE-0601 **Sub Name: IT Application & Data Security**

Time : 1 Hour **Maximum Marks: 15**

Section – A

3 x 2 = 6 Marks

Q.1 what do you understand by threat and malware. Discuss and explain various application security principals.

Q.2 how the same phases used in the traditional SDLC can be adapted to support the implementation of an information security project. Give the detailed analysis.

Q.3 Differentiate between Authentication and Authorization. Discuss how we are determining authentication vulnerability and protection from authentication vulnerability.

Section – B

3 x 3 = 9 Marks

Q.1 Alice and Bob wish to share private message, where each of them of two separate keys generated. What kind of strategy would you suggest to ensure confidentiality, key management, authentication and integrity for the conversation between Alice and Bob? Explain the strategy and also highlight the design issues related to the strategy proposed.

Q.2 what do you understand by input validation. Describe any two input validation vulnerability can be exploited by an attacker.

Q3 Write short note on any two:

a) Luring attack
b) Cookie replay attack
c) Brute force attack

Printed Pages: 01 **University Roll No.....**

First Mid-Term Examination, Odd Semester 2019-20

B.Tech CSE (Data Analytics), II, III

BCSE 0551 : Introduction to Business Analytics

Time : 1 Hour **Maximum Marks: 15**

Section – A

Attempt all the following questions 3 x 2 = 6 Marks

I. What is Analytics and why do we need it? Discuss how business decision usually taken?

II. "Business Analytics is multidisciplinary" Do you agree with statement? Provide reason for your agreement or disagreement. What is Optimization and what are the factors that cause hindrance in optimization?

III. Discuss how the following business get advantage from Analytics:

e. Google
f. Procter and Gamble
g. Obama Campaign
h. Netflix

Section – B

Attempt all the following questions 3 x 3 = 9 Marks

I. What are the main component of Business Analytics? Do discuss with examples?

II. What are the different classification of Analytics? Discuss then vividly and their advantages using case studies.

III. What is Data warehouse? Discuss different Schema involve in warehousing? What are the features of data Warehouse and explain them?

Printed Pages: 01

University Roll No.....

First Mid-Term Examination, Odd Semester 2019-20

B.Tech. (CS-IoT), Year - II, Sem. - III

BCSE 0651: Wireless Sensor Networks & IoT Standards

Time : 1 Hour

Maximum Marks: 15

Section – A

3 x 2 = 6 Marks

Attempt All questions.

I. What do you mean by Fault tolerance? Enlist few applications where low fault tolerance can also be sufficient. Enlist few applications which require high fault tolerance.

II. Enlist the different issues of a WSN? Which topology provides highest robustness and which topology requires the least number of links and I/O ports.

III. Calculate the transmission energy consumption and the energy consumed in receiving packets by a node with following parameters:
Initial Energy: 50mJ, E_{elec}=50nJ/bit, E_{amp}=100pJ/bit/m^2, R=5m, α=2, b=10bits

Section – B

3 x 3 = 9 Marks

Attempt All questions.

I. What are the different components of a sensor node? Briefly explain each of them.

II. Explain CSMA and SMAC protocols.

III. Mention the names of the different layers of OSI model in order. Describe data link layer and network layer.